Configuring Services for Hyper-V Virtualisation

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Lab Environment: CloudLabs VM (Browser-based)

Lab Objective

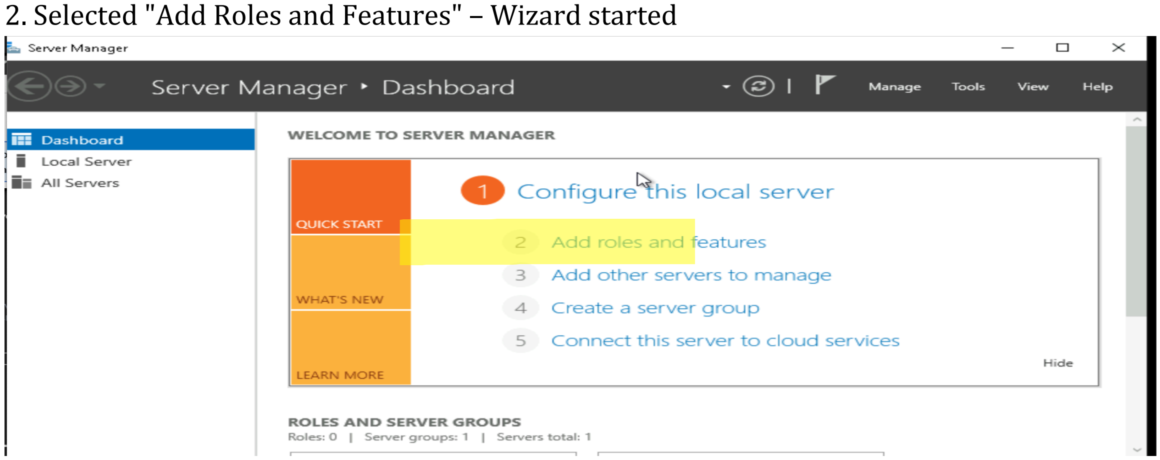
To configure Hyper-V virtualisation, DHCP Server, and Routing and Remote Access services on a Windows Server environment. This lab taught me how to turn a plain Windows Server into a virtualization host that can create, network, and manage virtual machines – a common setup in both corporate IT and security testing environments.

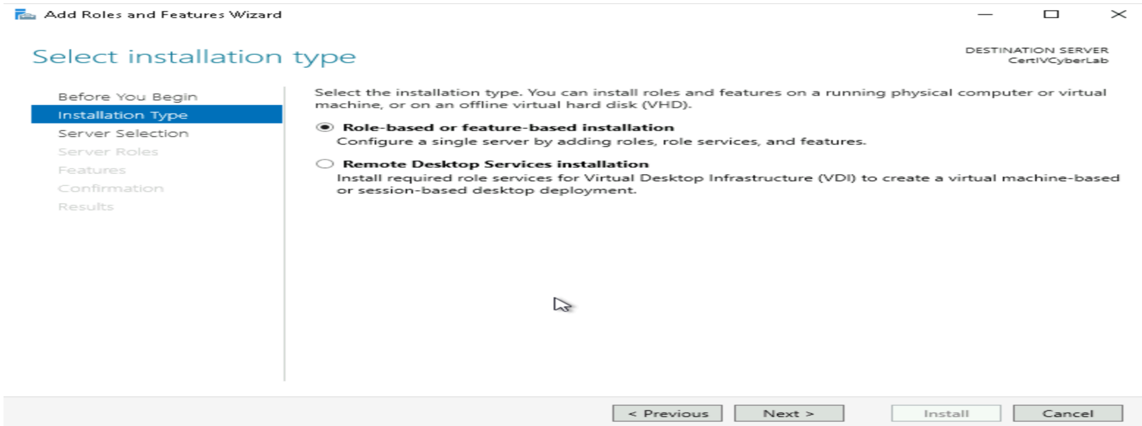
Hyper-V is a virtualization platform developed by Microsoft. It allows you to create and manage virtual machines on a Windows-based system. With Hyper-V you can run multiple operating systems on a single physical machine, enabling better resource utilization, isolation, and felixibility for testing and running different software environment.

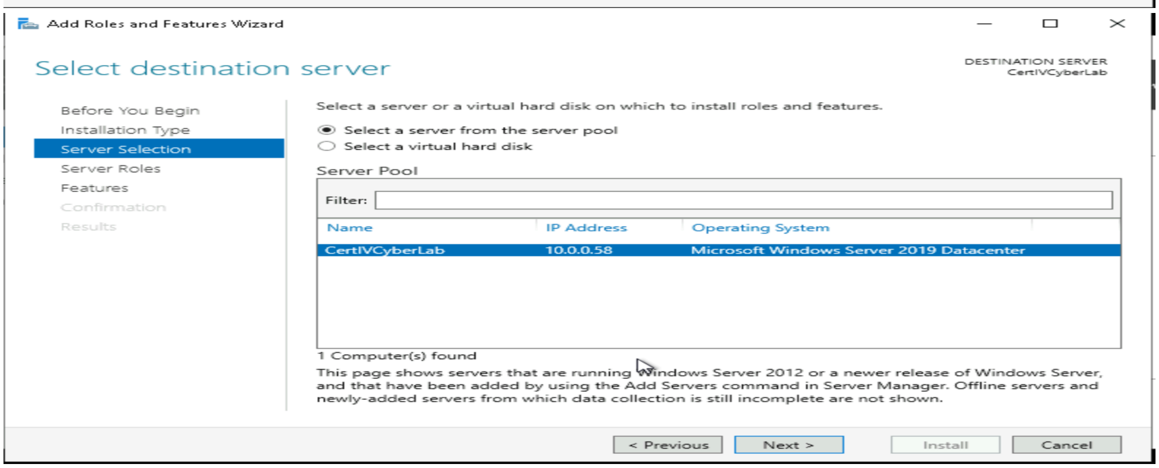
Task 1: Enabling Hyper-V on Windows

Steps Performed:

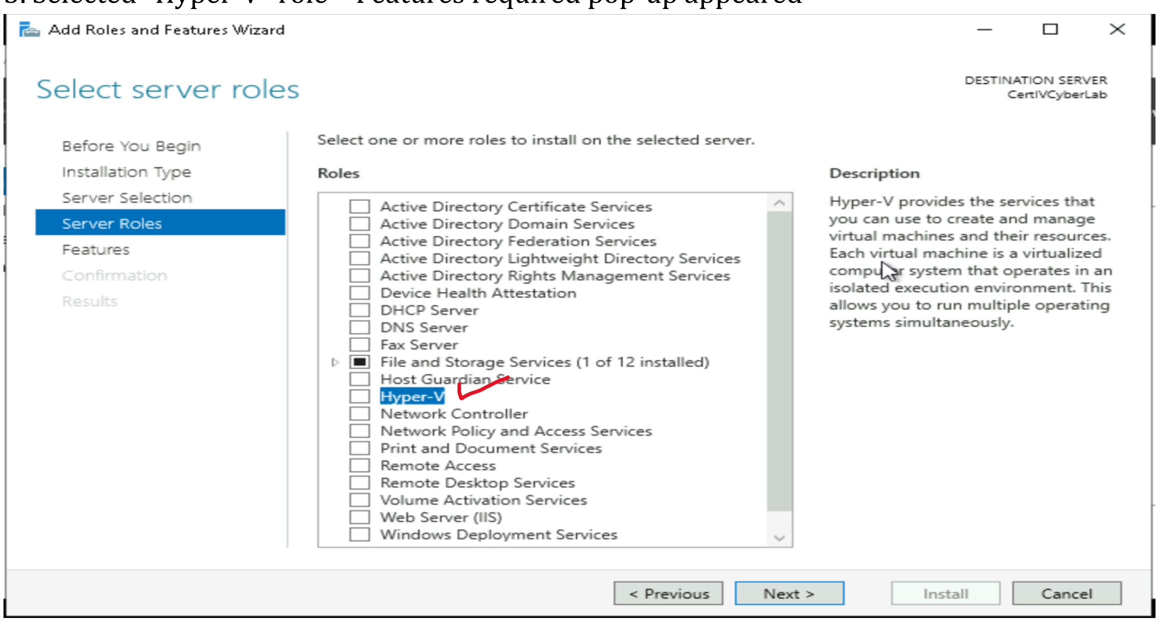
1. Opened Server Manager Dashboard – Server Manager loaded successfully
   2. Selected "Add Roles and Features" – Wizard started



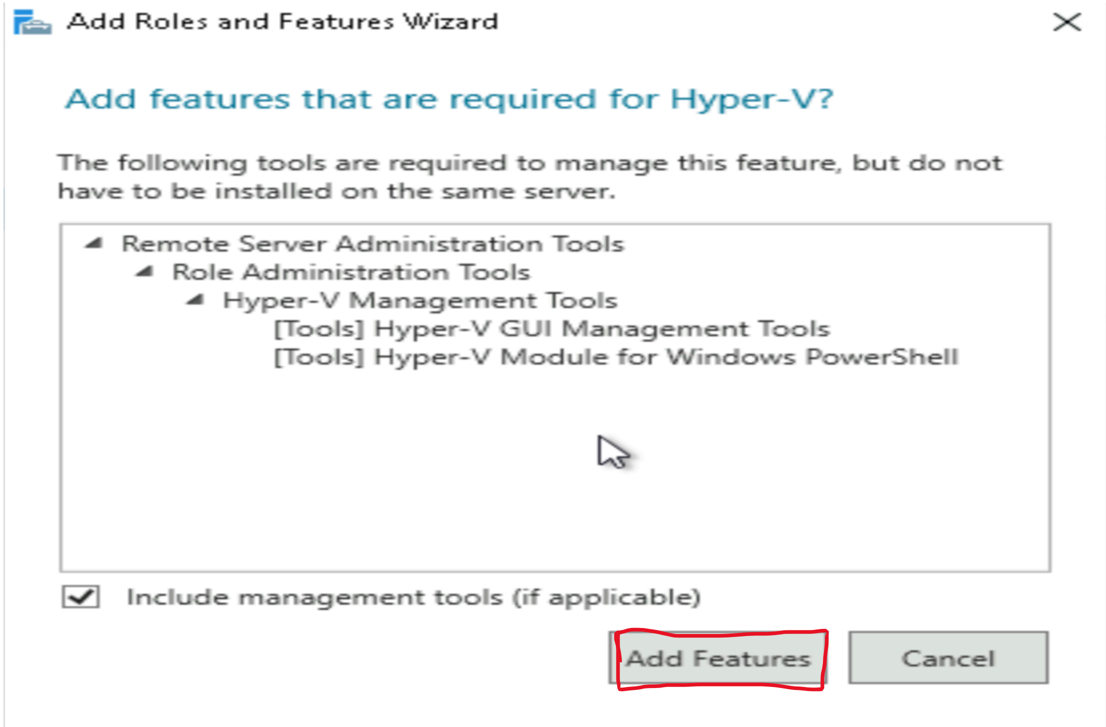


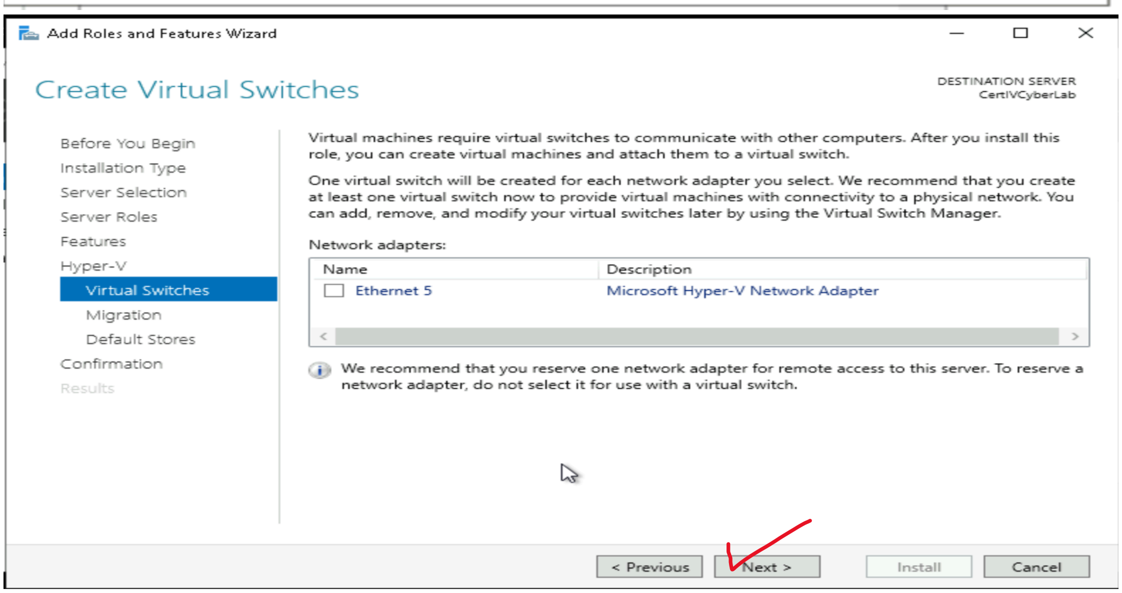


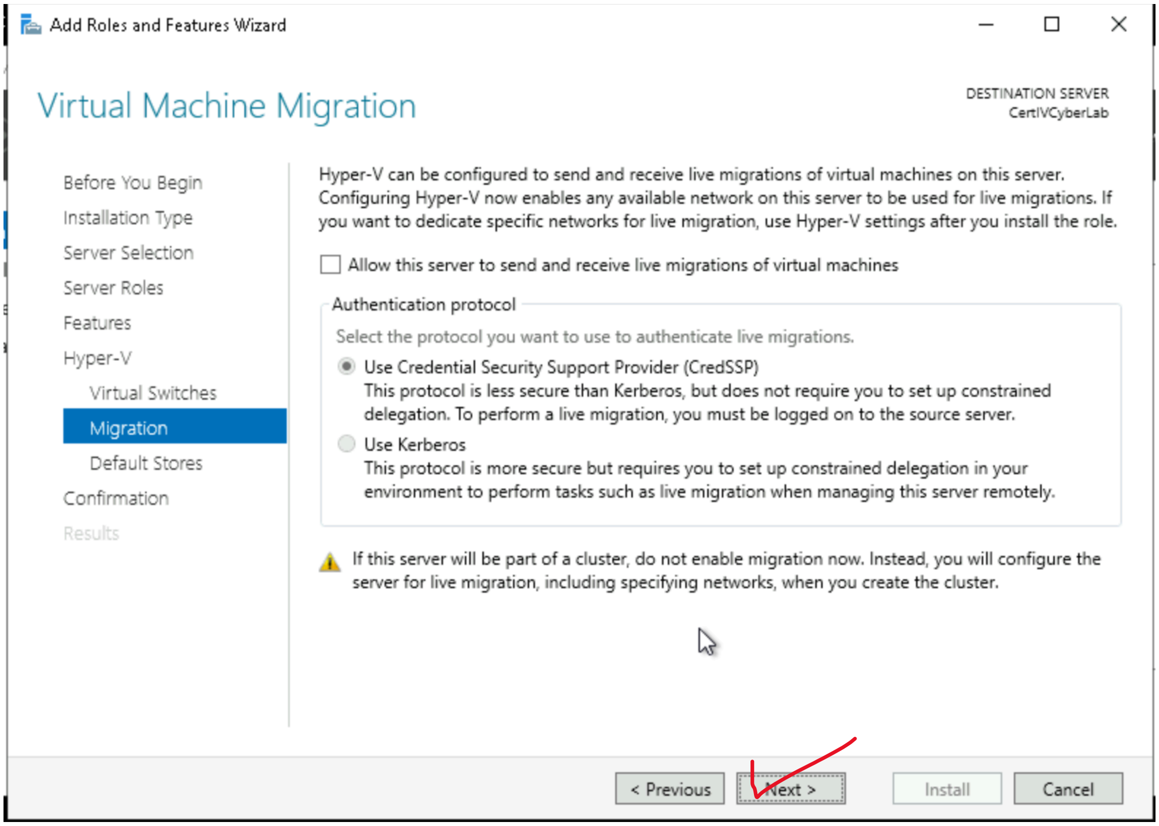
3. Selected "Hyper-V" role – Features required pop-up appeared

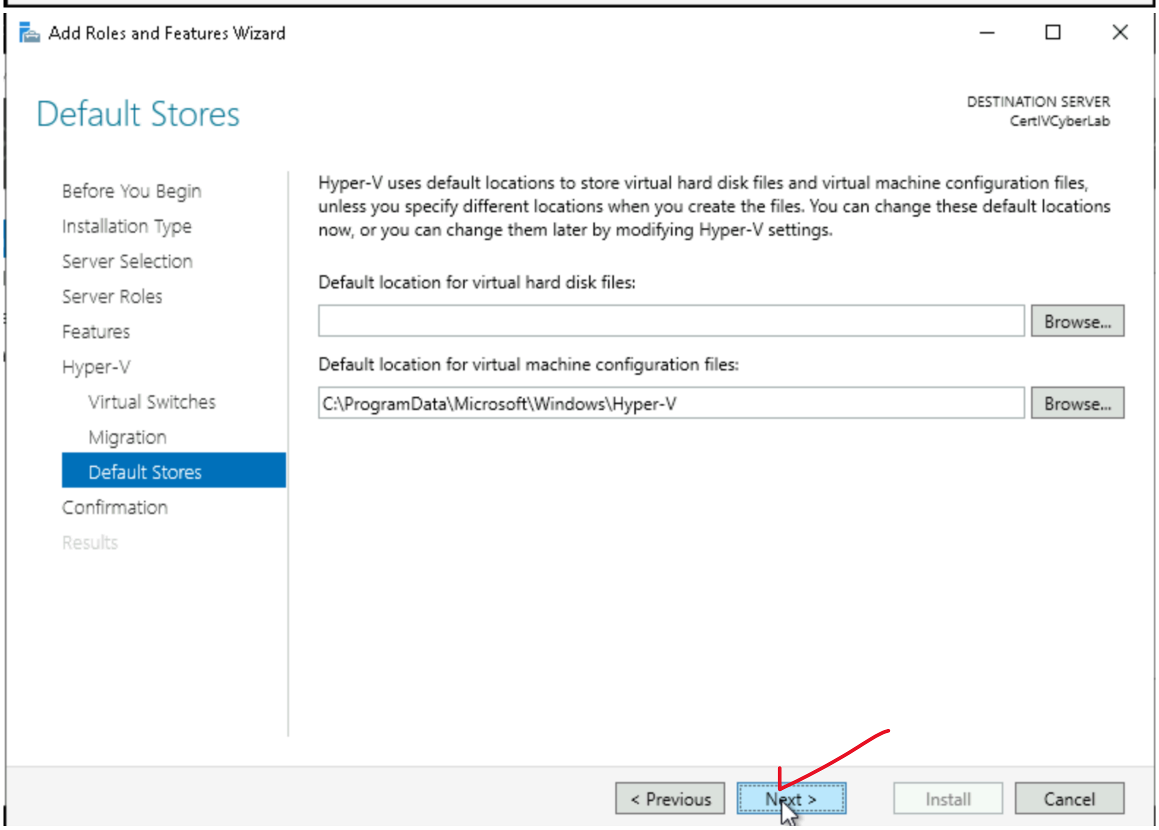


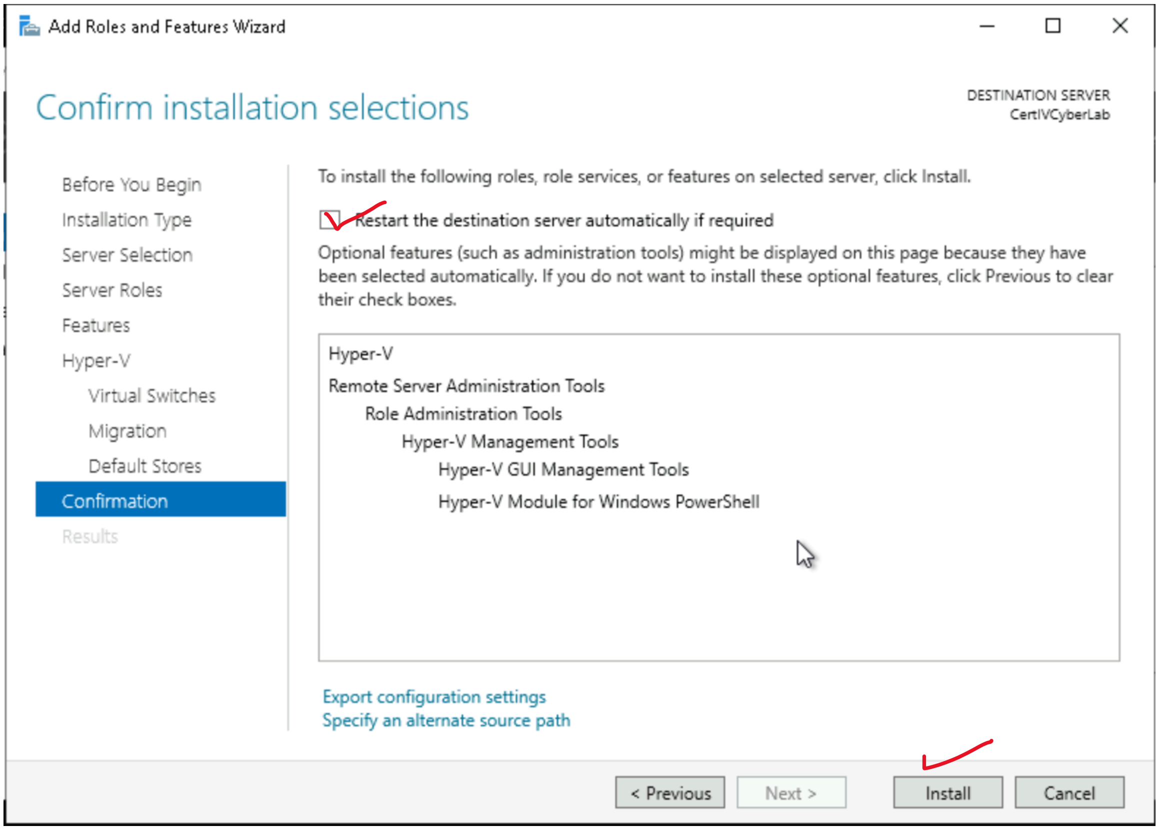
4. Added management tools – Features added


Add Roles and Features Wizard
Add features that are required for Hyper-V?
The following tools are required to manage this feature, but do not have to be installed on the same server.
Remote Server Administration Tools
Role Administration Tools
Hyper-V Management Tools
[Tools] Hyper-V GUI Management Tools
[Tools] Hyper-V Module for Windows PowerShell
Include management tools (if applicable)
Add Features
Cancel


Add Roles and Features Wizard
Create Virtual Switches
DESTINATION SERVER
CertIVCyberLab
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Hyper-V
Virtual Switches
Migration
Default Stores
Confirmation
Results
Virtual machines require virtual switches to communicate with other computers. After you install this role, you can create virtual machines and attach them to a virtual switch.
One virtual switch will be created for each network adapter you select. We recommend that you create at least one virtual switch now to provide virtual machines with connectivity to a physical network. You can add, remove, and modify your virtual switches later by using the Virtual Switch Manager.
Network adapters:
Name
Description
Ethernet 5
Microsoft Hyper-V Network Adapter
We recommend that you reserve one network adapter for remote access to this server. To reserve a network adapter, do not select it for use with a virtual switch.
< Previous
Next >
Install
Cancel

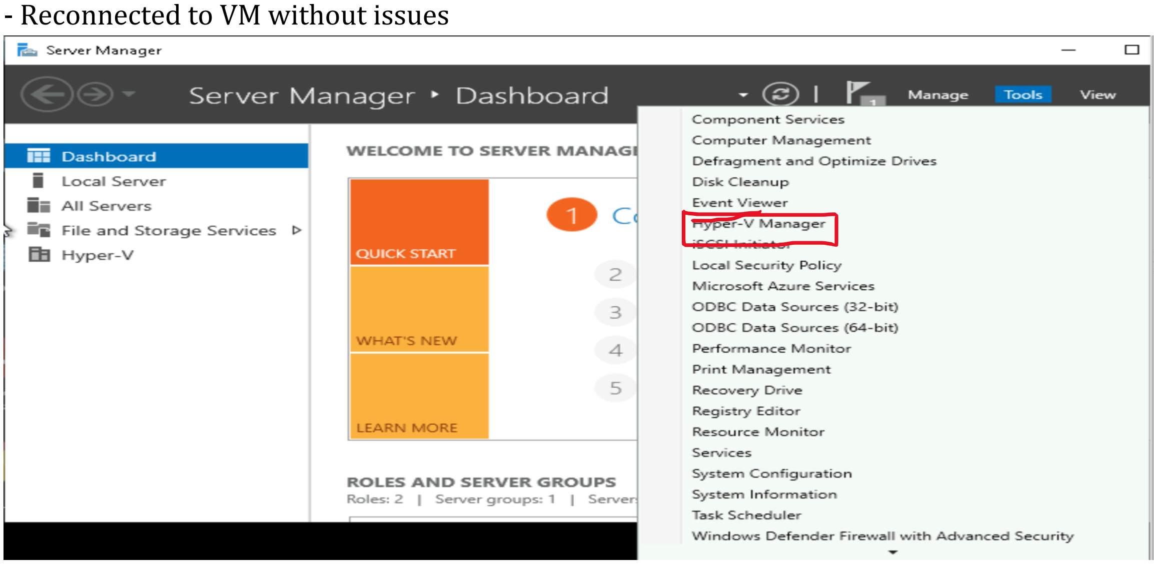

Add Roles and Features Wizard
Virtual Machine Migration
DESTINATION SERVER
CertIVCyberLab
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Hyper-V
Virtual Switches
Migration
Default Stores
Confirmation
Results
Hyper-V can be configured to send and receive live migrations of virtual machines on this server. Configuring Hyper-V now enables any available network on this server to be used for live migrations. If you want to dedicate specific networks for live migration, use Hyper-V settings after you install the role.
Allow this server to send and receive live migrations of virtual machines
Authentication protocol
Select the protocol you want to use to authenticate live migrations.
Use Credential Security Support Provider (CredSSP)
This protocol is less secure than Kerberos, but does not require you to set up constrained delegation. To perform a live migration, you must be logged on to the source server.
Use Kerberos
This protocol is more secure but requires you to set up constrained delegation in your environment to perform tasks such as live migration when managing this server remotely.
If this server will be part of a cluster, do not enable migration now. Instead, you will configure the server for live migration, including specifying networks, when you create the cluster.
< Previous
Next >
Install
Cancel


Add Roles and Features Wizard
Default Stores
DESTINATION SERVER
CertIVCyberLab
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Hyper-V
Virtual Switches
Migration
Default Stores
Confirmation
Results
Hyper-V uses default locations to store virtual hard disk files and virtual machine configuration files, unless you specify different locations when you create the files. You can change these default locations now, or you can change them later by modifying Hyper-V settings.
Default location for virtual hard disk files:
Browse...
Default location for virtual machine configuration files:
C:\ProgramData\Microsoft\Windows\Hyper-V
Browse...
< Previous
Next >
Install
Cancel



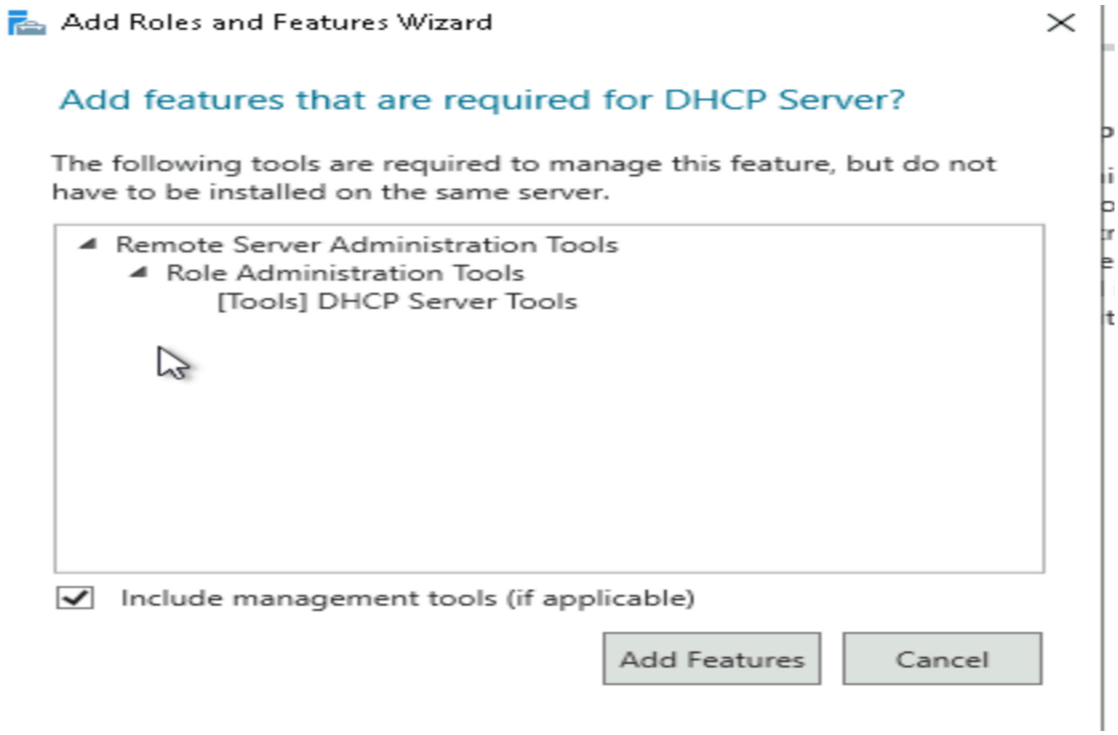
5. Completed installation with auto-restart – Server restarted automatically
6. Verified installation – Hyper-V Manager appeared in Tools menu

Verification:
- Hyper-V Manager visible in Server Manager → Tools menu
- Server restarted successfully
- Reconnected to VM without issues

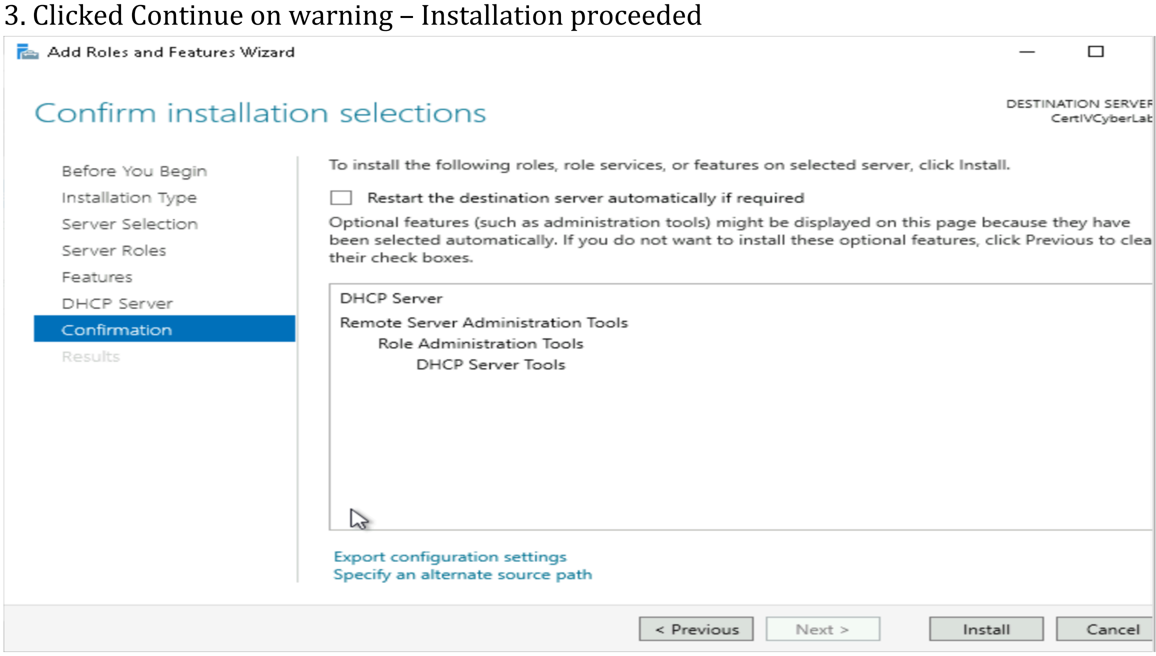


Task 2: Installing DHCP Server

Steps Performed:
1. Opened Add Roles and Features Wizard – Wizard started
2. Selected "DHCP Server" role – Management tools added



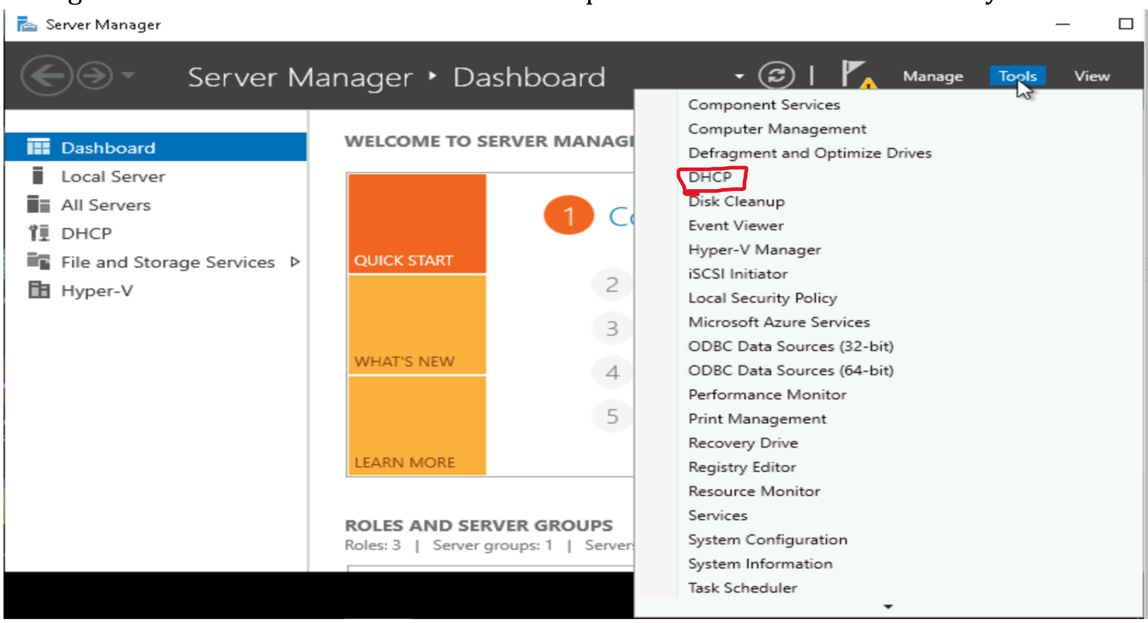
3. Clicked Continue on warning – Installation proceeded



4. Completed installation – DHCP Server installed
5. Verified installation – DHCP appeared in Tools menu

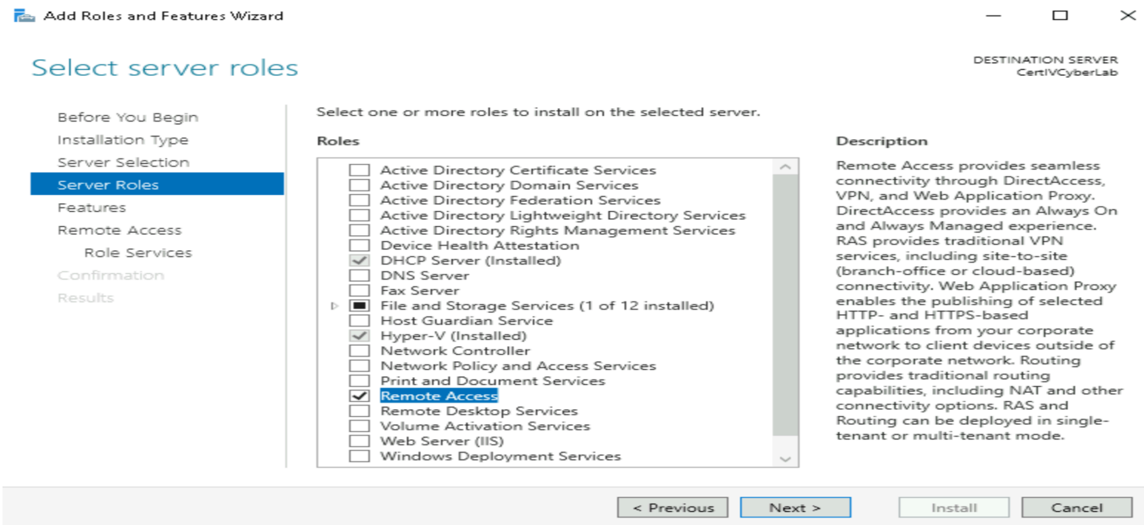
Verification:
- DHCP Server visible in Server Manager → Tools menu
- Installation completed without errors

Notes:
The warning about static IP addressing appeared as expected in the lab instructions. This was ignored as the CloudLabs environment requires DHCP for RDP connectivity.

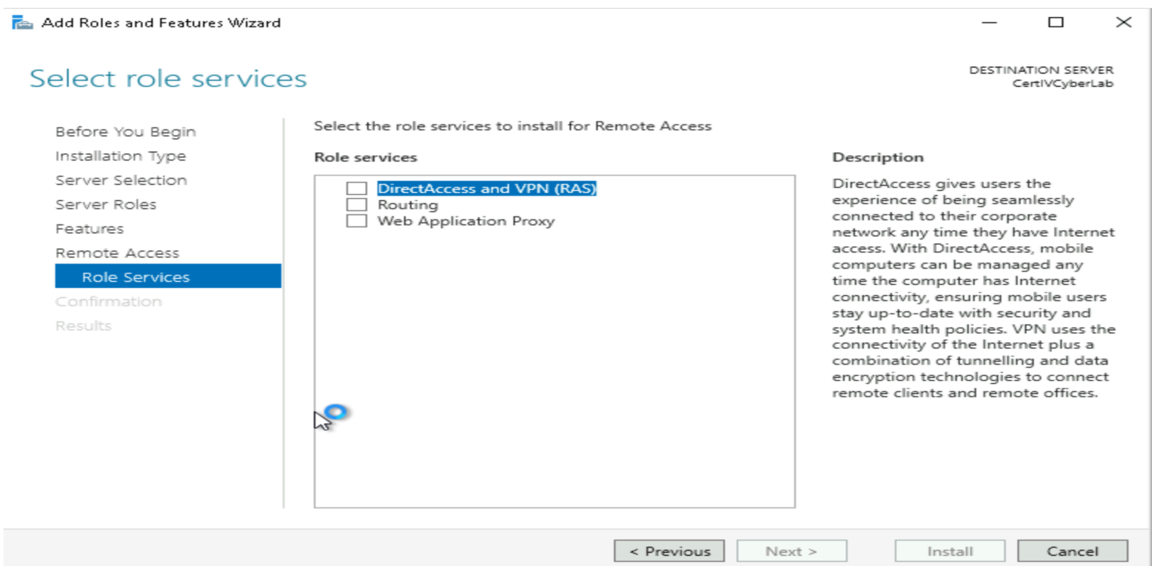


Task 3: Enabling Routing and Remote Access

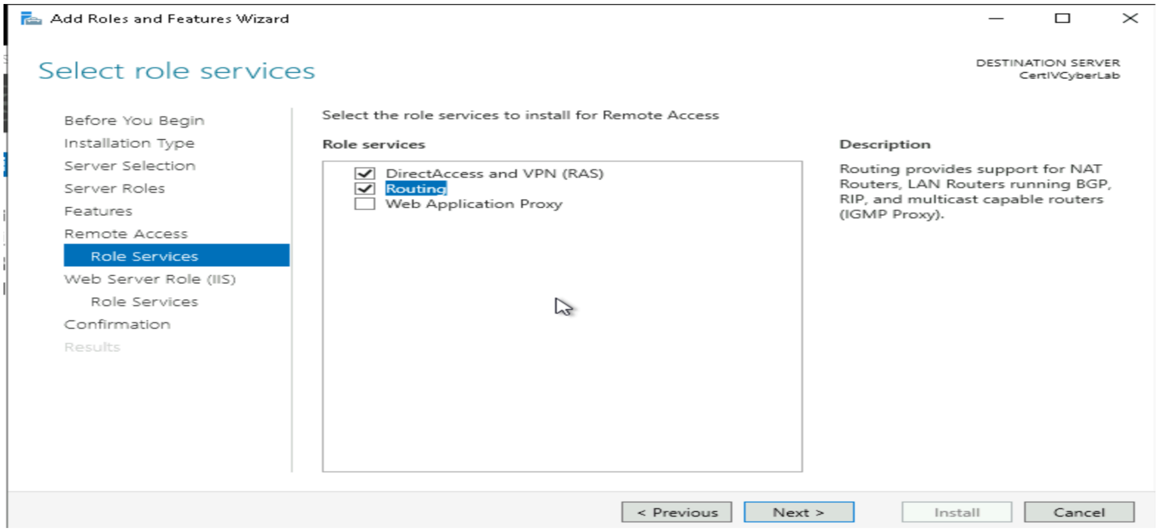
Steps Performed:
1. Opened Add Roles and Features Wizard – Wizard started
2. Selected "Remote Access" role – Role services page appeared



3. Selected "DirectAccess and VPN (RAS)" – Features pop-up appeared

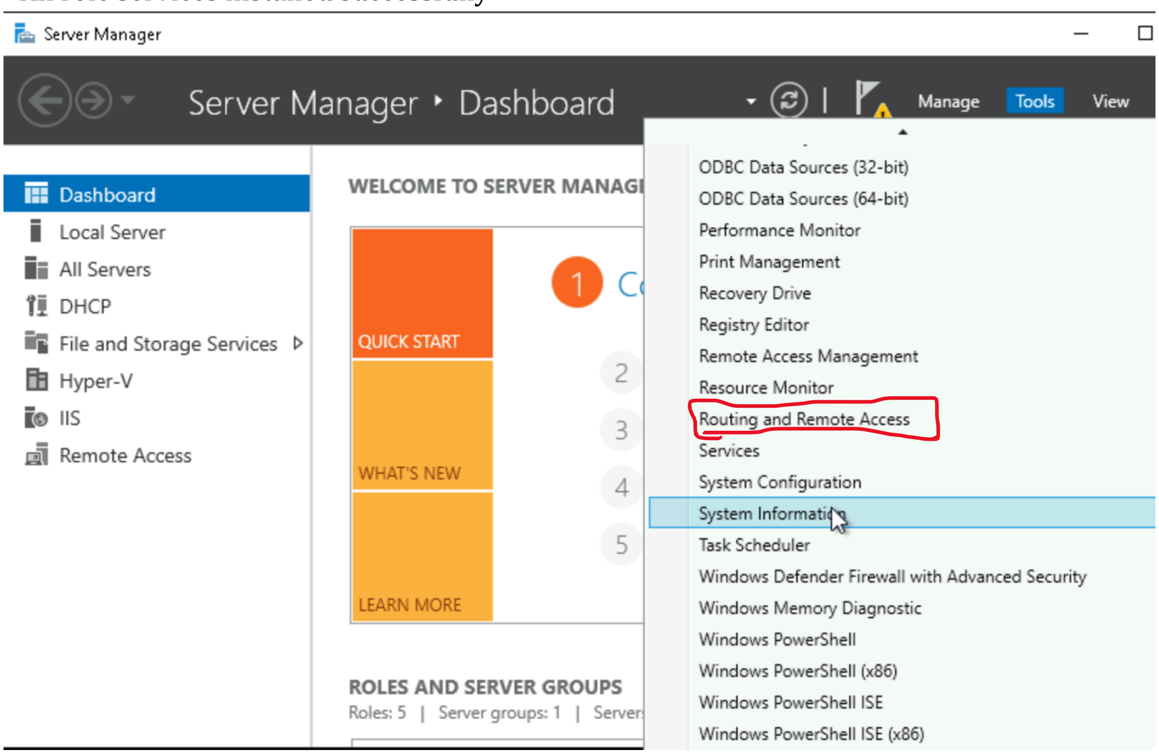


4. Added management tools – Features added
5. Selected "Routing" role service – Routing selected



6. Completed installation – Remote Access installed
7. Verified installation – Routing and Remote Access in Tools menu

Verification:
- Routing and Remote Access visible in Server Manager → Tools menu
- All role services installed successfully



Final Verification

All three services are now visible in the Server Manager Tools menu:
- Hyper-V Manager – Installed

- DHCP – Installed
- Routing and Remote Access – Installed

Challenges Encountered

After Hyper-V installation, the VM took about 3 minutes to restart. I waited and clicked Reconnect in the browser. No major challenges were encountered, and all steps worked as described.

Reflection

I learned how to add Windows Server roles and features using Server Manager. I understand that nested virtualisation allows running VMs inside other VMs. I now know how to verify service installation using the Tools menu.